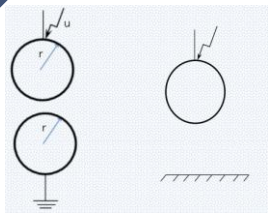
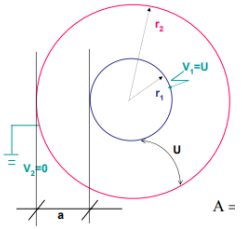
EE4047 – High Voltage Techniques I

Continuation to:
Electrical Breakdown in Gases





Week 11.2 – Thursday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

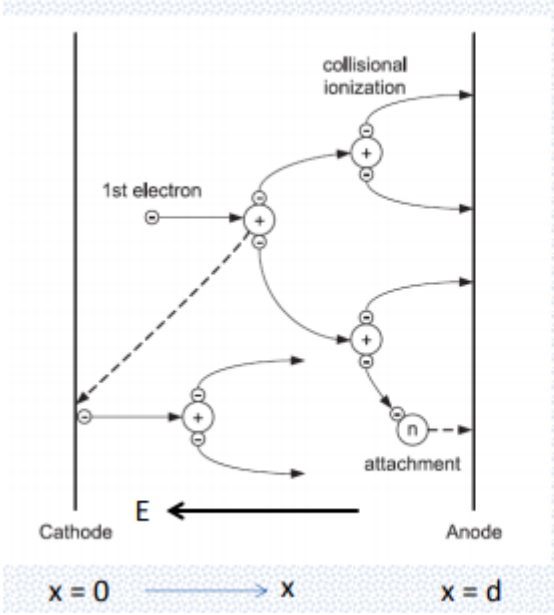
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OUTLINE

4 Continuation to: Discharge theorems

1 DISCHARGE THEOREMS

1. Townsend Theorem



γ: is the number of electrons removed from the cathode by one positive ion striking to cathode.

$$\gamma: \frac{1}{50}, \dots, \frac{1}{5000}$$

If γ =1/50, one electron can be removed from the cathode when 50 positive ions strike to it.

1 DISCHARGE THEOREMS

1. Townsend Theorem

$\gamma = 0.02 \sim 0.00002$

γ depends on the material of the cathode and the type os gaseous insulator.

Material of cathode	Gas	
	Air	Neon
Al	0.035	0.1
Cu	0.025	0.065
Fe	0.02	0.06

1 DISCHARGE THEOREMS

1. Townsend Theorem

At a distance x = d,

$(e^{\alpha d} - 1)$ positive ions are produced by collison

$\gamma(e^{\alpha d} - 1)$ electrons are produced

Number of electrons removed from the cathode by one positive ion

1 DISCHARGE THEOREMS

1. Townsend Theorem

$$\gamma\left(e^{\alpha d}-1\right)=1$$

If this condition is fulfilled, at least one electron is removed from the cathode and discharge can continue.

This is also called as; **condition of self-sustainability of discharge.**

1 DISCHARGE THEOREMS

1. Townsend Theorem

Equation for breakdown voltage in uniform fields by Townsend's theory (Paschen Equation)

$$U_b = \frac{B.p.d}{\ln\left[\frac{A.p.d}{\ln(1+\frac{1}{\gamma})}\right]}$$

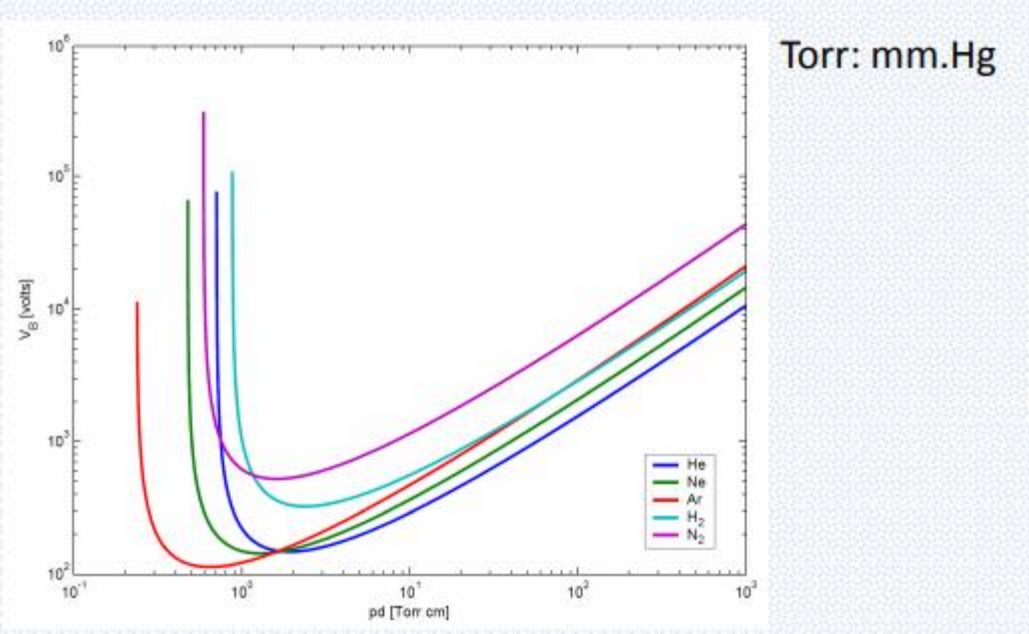
$$U_b = \mathrm{f(p.d)}$$

Breakdown voltage depends on multiplication of pressure and electrode gap

1 DISCHARGE THEOREMS

1. Townsend Theorem

Breakdown voltage curve (Paschen curve)



1 DISCHARGE THEOREMS

1. Townsend Theorem

The critical (p.d) value that makes the breakdown voltage minimum

$$(p.d)_{cr} = \frac{e.\ln(1+\frac{1}{\gamma})}{A}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

The critical (p.d) value that makes the breakdown voltage minimum

$$U_b[(p.d)_{cr}] = U_{b,min} = \frac{B.(p.d)_{cr}}{\ln\left[\frac{A.(p.d)_{cr}}{\ln(1+\frac{1}{\gamma})}\right]}$$

$$U_{b,min} = \frac{B.\frac{e.\ln(1+\frac{1}{\gamma})}{A}}{\ln\left[\frac{A.\frac{e.\ln(1+\frac{1}{\gamma})}{A}}{\ln(1+\frac{1}{\gamma})}\right]} = B.\underbrace{\frac{e.\ln(1+\frac{1}{\gamma})}{A}}_{(p.d)_{cr}}$$

$$U_{b,min} = B.(p.d)_{cr}$$

1 DISCHARGE THEOREMS

1. Townsend Theorem

The critical (p.d) value that makes the breakdown voltage minimum

$$E = {}^{U}/_{d}$$

$$E_b = {}^{U_b}/_{d}$$

$$E_b = \frac{B.p}{\ln\left[\frac{A.p.d}{\ln(1+\frac{1}{\gamma})}\right]}$$

DISCHARGE THEOREMS

1. Townsend Theorem

Generalized Pasched Equation

$A \rightarrow A.{}^{T}/_{T_0}$
$B \rightarrow B.{}^{T}/_{T_0}$

$$U_b = \frac{B.\frac{T}{T_0}.p.d}{\ln\left[\frac{A.\frac{T}{T_0}.p.d}{\ln(1+\frac{1}{\gamma})}\right]}$$

- $T_0 = 273 + 20°C = 293°K$
- $T = 273 + \theta°C$: gas temperature in $°K$
- p: gas pressure in mmHg
- A, B: gas constants
- γ: number of electrons removed from the cathode by positive ions

1 DISCHARGE THEOREMS

Streamer Theory

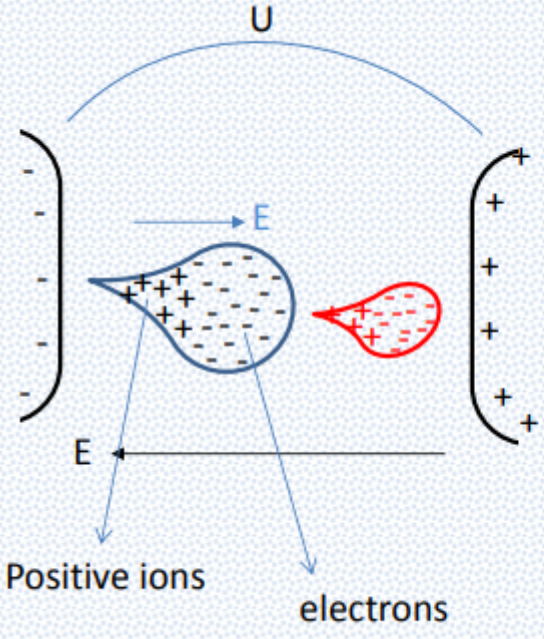
This theory explains electrical discharges in big gaps and high pressure.

p.d>500 mmHg.cm

The Streamer Theory predicts the development of a spark discharge directly from a single avalanche.

DISCHARGE THEOREMS

Streamer Theory



the electrons advance rapidly, the positive ions are left behind in a relatively slow-moving tail

Electric field between the avalanche and the anode is big

DISCHARGE THEOREMS

Streamer process

• The space charge produced in the avalanche causes sufficient distortion of the electric field that those free electrons move towards the avalanche head, and in so doing generate further avalanches in a process that rapidly becomes cumulative.
• As the electrons advance rapidly, the positive ions are left behind in a relatively slow-moving tail
• The field will be enhanced in front of the head. Just behind the head the field between the electrons and the positive ions is in the opposite direction to the applied field and hence the resultant field strength is less.
• Again between the tail and the cathode the field is enhanced.
• Due to the enhanced field between the head and the anode, the space charge increases, causing a further enhancement of the field around the anode.
• The process is very fast and the positive space charge extends to the cathode very rapidly resulting in the formation of a streamer.

DISCHARGE THEOREMS

Breakdown voltage by Streamer Theory

$$U_b = 5.27 \times 10^{-7} \frac{\alpha . d . e^{\alpha d}}{\sqrt{\frac{d}{p}}}$$

U_b: V

α: 1/cm

d: cm

p: mmHg

d: electrode gap [cm] (electrode seperation)

p: gas pressure [mmHg]

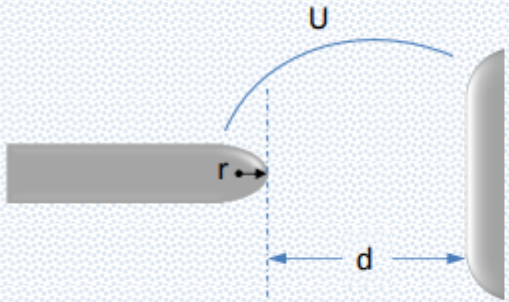
α: ionization coefficient of the eletron [1/cm]

U_b: breakdown voltage [V]

$$E_b = \frac{U_b}{d} = 5.27 \times 10^{-7} \frac{\alpha . e^{\alpha d}}{\sqrt{\frac{d}{p}}}$$

DISCHARGE THEOREMS

Streamer Theory for non-uniform electric field



r: curvature radius

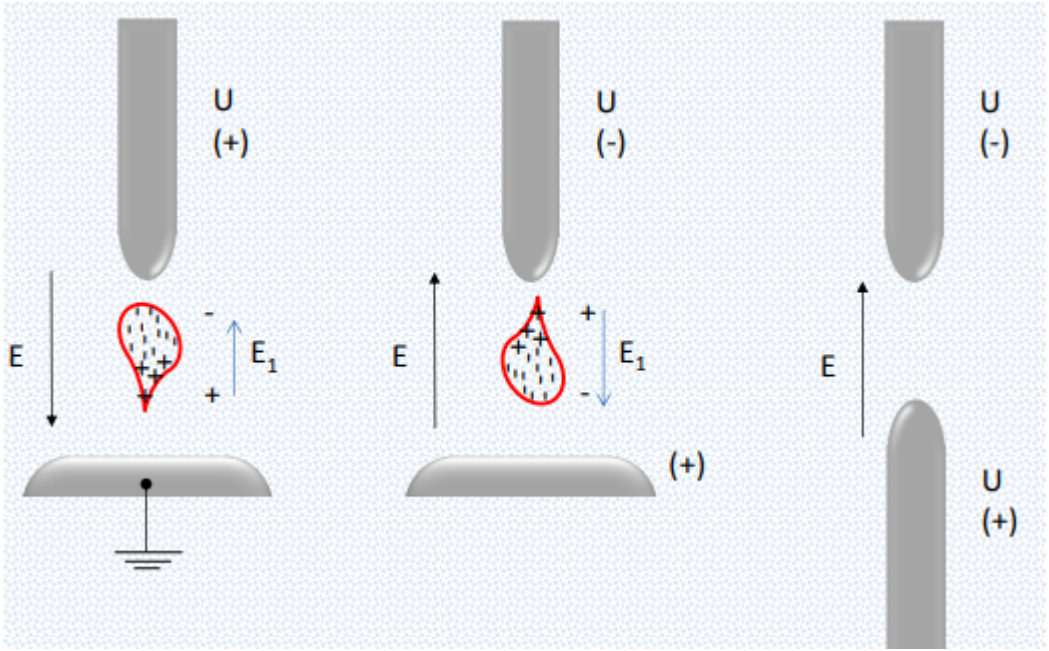
d: electrode gap

Non-uniform field:

$$\frac{d}{r} > 40 \rightarrow d > 40\,r$$

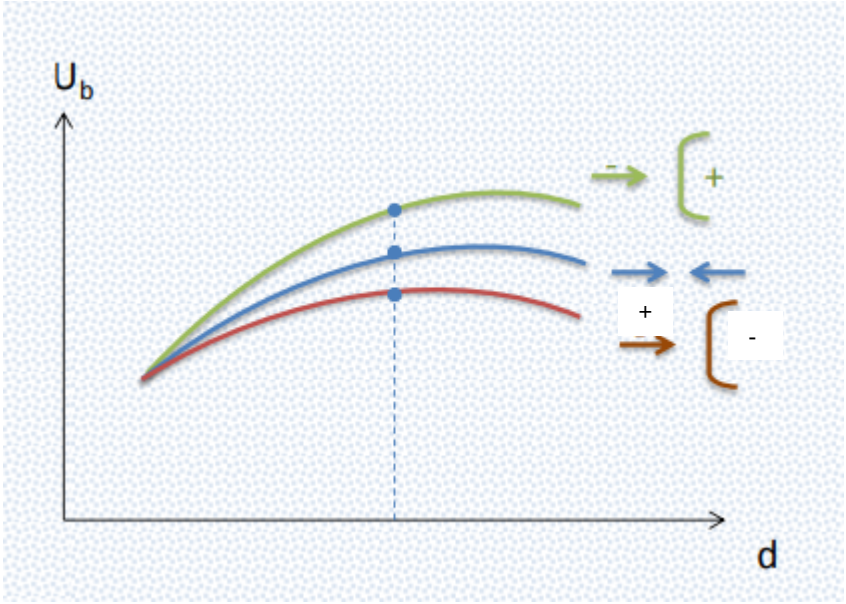
DISCHARGE THEOREMS

Different non-uniform electric field cases



DISCHARGE THEOREMS

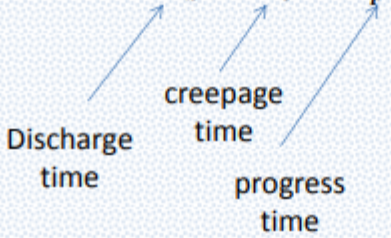
Electrode gap – breakdown voltage



1 DISCHARGE THEOREMS

Discharge time

$$t_b = t_\sigma + t_p$$



Creepage time: time required for an electron to appear after the moment which voltage applied

Progress time: time required for such an electron appeared in the gap to lead a breakdown after the moment it appeared

EE4047 – High Voltage Techniques I

New Topic:
Corona Discharges

Week 11.2 – Thursday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

18.05.2023

OUTLINE

1 Definition

2 Characteristics Properties

3 Mechanism of Corona

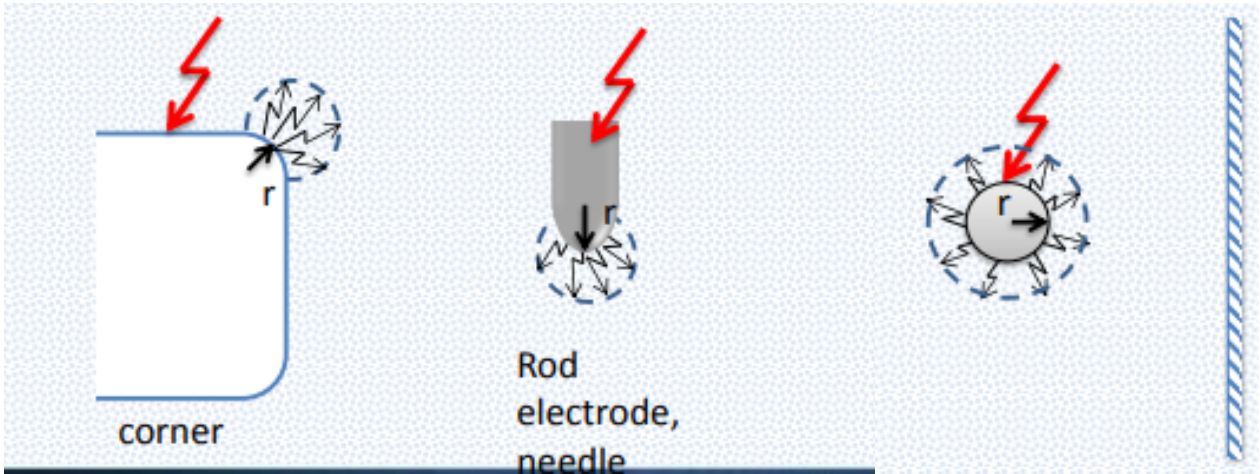
4 Power Loss due to Corona

1 DEFINITION

Corona

Literal meaning of corona is ‘’light crown’’.

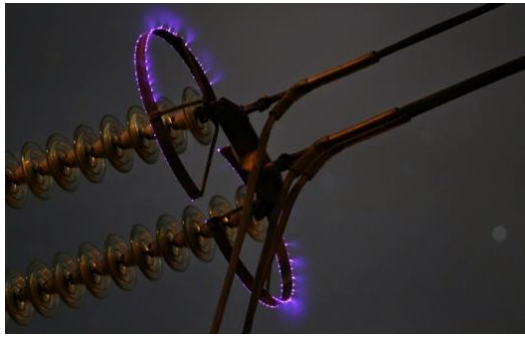
Physical meaning of corona is partial discharge occuring around regions with small curvature radius (sharp edges, corners, needles, ...)



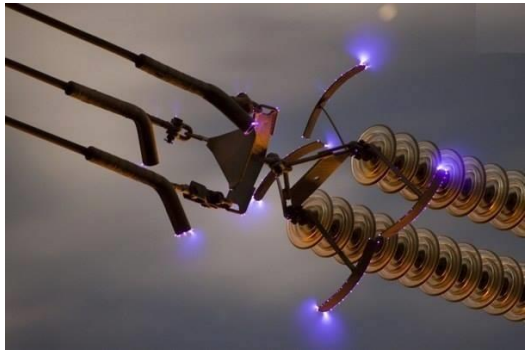
1 DEFINITION

Corona

In a uniform electric field, a gradual increase in voltage across a gap produces a breakdown of the gap in the form of a spark without any preliminary discharges (direct flashover / breakdown without corona dishcarge).



On the other hand, if the field is non-uniform, an increase in voltage will first cause a localised discharge in the gas to appear at points with the highest electric field intensity, namely at sharp points or where the electrodes are curved or on transmission line conductors.



This form of discharge is called a **corona discharge** and can be observed as a bluish luminance.

2 CHARACTERISTIC PROPERTIES

Corona has a specific,

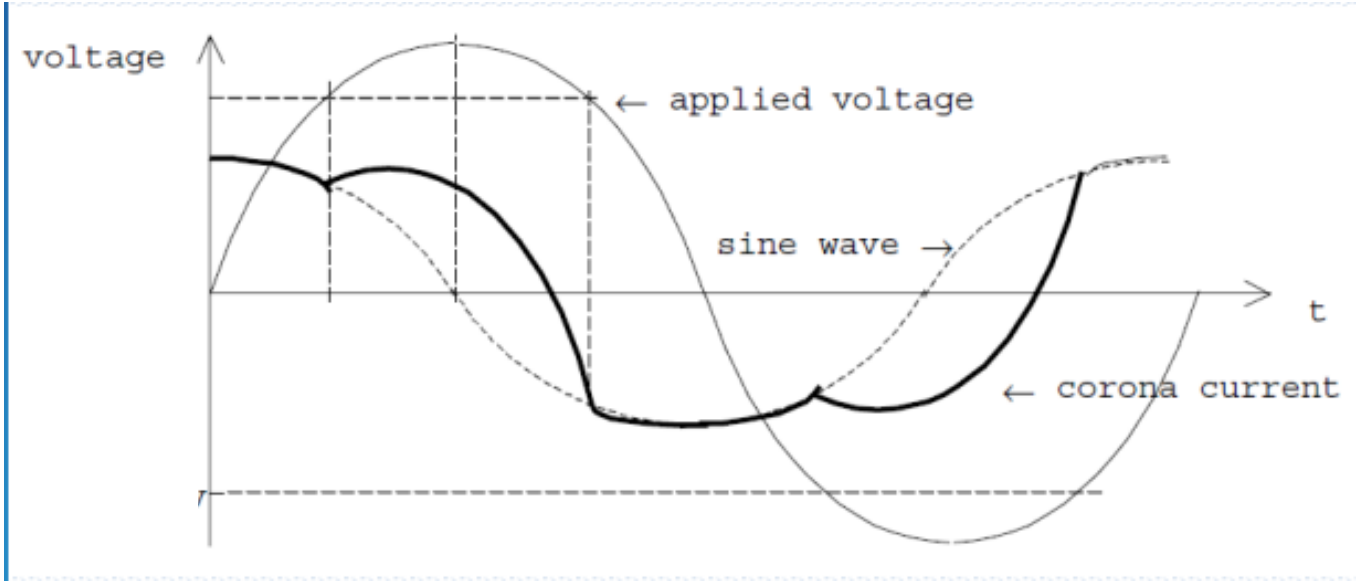
- Light
- Sound (hummin noise)
- Smell (O_3 - ozone gas production)

Corona is more common in transmission lines & equipment of 245 kV and above.

Corona causes power loss, production of high frequency noise, interference with radio and TV, and production of ozone gas.

2 CHARACTERISTIC PROPERTIES

Waveform of a corona current



2 CHARACTERISTIC PROPERTIES

The shunt current in a line is almost purely capacitive under normal conditions, and leads the applied voltage by 90°, and there is no power loss in the line under no-load conditions.

When the applied voltage is increased and corona is formed, the air is rendered conducting and power loss occurs.

The shunt current would no longer be leading the voltage by 90°. Thus the current waveform would consist of two components.

The lossy component would be non-sinusoidal and would occur only when disruptive critical voltage is exceeded in either polarity.

3 MECHANISM OF CORONA

The stress surrounding the conductor is a maximum at the conductor surface itself, and decreases rapidly as the distance from the conductor increases.
Thus, when the stress has been raised to critical value immediately surrounding the conductor, ionisation would commence only in this region and the air in this region would become conducting.
The effect is to increase the effective conductor diameter while the voltage remains constant.

This has **two effects**:

1. An increase in the effective sharpness of the conductor would **reduce the stress** outside this region,
2. This would cause a reduction of the effective spacing between the conductors leading to an **increase in stress**.

Depending on which effect is stronger, the stress at increasing distance can either increase or decrease. If the stress is made to increase, further ionisation would occur and flashover is inevitable.

3 MECHANISM OF CORONA

Under ordinary conditions, the breakdown strength of air can be taken as 30 kV/cm.

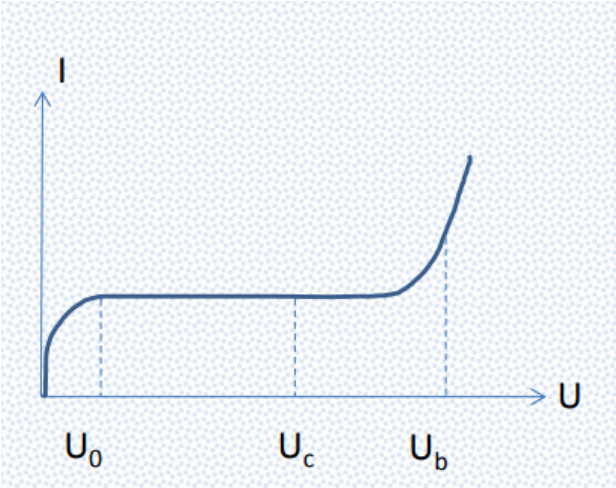
Corona will of course be affected by the physical state of the atmosphere.

In stormy weather, the number of ions present is generally much more than normal, and corona will then be formed at a much lower voltage than in fair weather.

This reduced voltage is generally about 80% of the fair weather voltage.

3 MECHANISM OF CORONA

Current voltage characteristic



POWER LOSS DUE TO CORONA

Power loss due to Corona

The formation of corona is associated with a loss of power.

This loss will have a small effect on the efficieny of the line, but will not be of sufficient importance to have any appreciable effect on the voltage regulation.

The more important effect is the radio interference.

POWER LOSS DUE TO CORONA

Corona Voltage

$$U_c = U_0 \times m \times \delta \times \left(1 + \frac{0.301}{\sqrt{r \times \delta}}\right)$$

Irregularity factor

Corona voltage

Corona inception voltage

curvature radius

Relative air density

m= 1 – 0.8

$$\delta = \frac{p}{p_0} \cdot \frac{T_0}{T} = \frac{p\,[mmHg]}{760[mmHg]} \cdot \frac{273 + 20°C}{273 + \theta°C} = 0.386 \frac{p[mmHg]}{273 + \theta°C}$$

POWER LOSS DUE TO CORONA

For an overhead line

- Single phase:
- $U_c = 21.2 \times m \times r \times \delta \times \left(1 + \frac{0.301}{\sqrt{r \times \delta}}\right) \times \ln \frac{d}{r}$

kVeff

$\frac{30}{\sqrt{2}}$ kV

Electrode gap

- Three phase:
- $U_c = \sqrt{3} \times 21.2 \times m \times r \times \delta \times \left(1 + \frac{0.301}{\sqrt{r \times \delta}}\right) \times \ln \frac{d}{r}$

POWER LOSS DUE TO CORONA

Corona loss (Peek's formula)

$$P_c = \frac{243}{\delta} \times (f + 25) \times \sqrt{\frac{r}{d}} \times (U - U_0)^2 \times 10^{-5} \quad \frac{kW}{km.phase}$$

4 POWER LOSS DUE TO CORONA

Corona can be mainly minimised by,

- increasing the diameter of conductor,
- by bundling the conductor,
- by using corona ring.

END OF THE LESSON

Any questions?

Check out the following videos on Youtube:

Electrical Corona Effect: https://www.youtube.com/watch?v=m-jeQgXAqJA

Flare Corona Discharge: https://www.youtube.com/watch?v=C0Fry6ktu4w

Corona Effect Recorded with Special Camera: https://www.youtube.com/watch?v=5bKcJegKfBo